WAS Experiment 03

Aim: Burp Repeater and Intruder

Learning Objective: To demonstrate the use of Burp Repeater and Burp Intruder for analyzing and testing HTTP requests in a controlled web application security testing environment.

Learning Objectives:
After completing this experiment, students will be able to:

- Understand the purpose of Burp Repeater.
- Understand the purpose of Burp Intruder.
- Capture HTTP requests using Burp Proxy.
- Send captured requests to Repeater.
- Modify and resend HTTP requests.
- Analyze HTTP responses using Repeater.
- Configure Intruder positions.
- Perform controlled automated request testing using Intruder.
- Compare responses generated by different inputs.
- Understand the role of Repeater and Intruder in web application security testing.

Tools: Kali Linux, Burp Suite, Web Browser - Firefox, a local or authorized web application and Internet connection.

Theory:

Introduction

During web application security testing, security testers often need to modify HTTP requests and observe how the application responds to different inputs.
Burp Repeater and Burp Intruder are two important tools in Burp Suite that assist with this type of testing.
Repeater is primarily used for manual, controlled testing of individual HTTP requests, while Intruder is used to automate the process of sending multiple variations of a request.

Burp Repeater

Burp Repeater allows a tester to capture an HTTP request, modify it, and resend it repeatedly to the target application.
It is useful for:

- Testing individual parameters.
- Modifying HTTP headers.
- Changing request values.
- Comparing server responses.
- Manually investigating application behavior.
- Testing input validation.

The basic process is:
Capture Request
↓

Send to Repeater
↓
Modify Request
↓
Send Request
↓
Analyze Response

Burp Intruder

Burp Intruder is used to automate the process of sending multiple requests with different input values. It can be used for authorized security testing such as:

- Input validation testing.
- Parameter fuzzing.
- Controlled authentication testing.
- Testing application responses to different inputs.

The basic process is:

Capture Request
↓
Send to Intruder
↓
Select Positions
↓
Configure Payloads
↓
Start Attack
↓
Analyze Responses

Procedure:

Step 1: Start Burp Suite

Start Kali Linux and launch Burp Suite.

burpsuite

Select Temporary Project and continue with the default configuration.

Expected Output:

Burp Suite starts successfully and the main interface is displayed.

Step 2: Configure the Browser

Configure Firefox to use Burp Suite as the proxy.

Proxy Address: 127.0.0.1

Proxy Port: 8080

Expected Output:

Browser traffic is routed through Burp Suite.

Step 3: Open the Target Application

Open an authorized web application in Firefox.

For example:

http://localhost/

Expected Output:

The target web application loads successfully.

Step 4: Capture an HTTP Request

Use the application to generate an HTTP request.

In Burp Suite, open:

Proxy → Intercept

Capture the request generated by the browser.

Expected Output:

The HTTP request is displayed in Burp Suite.

Burp Repeater

Step 5: Send the Request to Repeater

Right-click the captured request and select:

Send to Repeater

Open:

Repeater

The captured HTTP request is displayed.

Expected Output:

The request is available for manual modification and testing.

Step 6: Modify the Request

Change a parameter or other appropriate part of the request.

For example, modify a parameter value:

parameter=value1

to:

parameter=value2

Click Send.

Expected Output:

Burp Suite sends the modified request to the server.

Step 7: Analyze the Response

Observe the response returned by the server.

Compare:

- HTTP status code
- Response length
- Response content
- Headers
- Application behavior

Expected Output:

The response corresponding to the modified request is displayed.

Burp Intruder

Step 8: Send the Request to Intruder

Return to the captured HTTP request.

Right-click the request and select:

Send to Intruder

Open:

Intruder

Expected Output:

The request is displayed in the Intruder interface.

Step 9: Select Payload Positions

In the Positions section, identify the parameter that will be tested.

Burp Suite uses markers to identify the selected position.

For example:

username=§test§

The selected value between § symbols represents the payload position.

Expected Output:

The parameter to be tested is marked as the payload position.

Step 10: Configure Payloads

Open the Payloads section and provide a small set of authorized test values.

Example:

test1

test2

test3

admin

student

Expected Output:

The payload list is configured successfully.

Step 11: Start the Intruder Attack

Start the attack using the configured payloads.

Burp Intruder sends requests using the different payload values.

Expected Output:

Multiple requests are generated and their responses are displayed in the results table.

Step 12: Analyze Intruder Results

Compare the responses generated for different payloads.

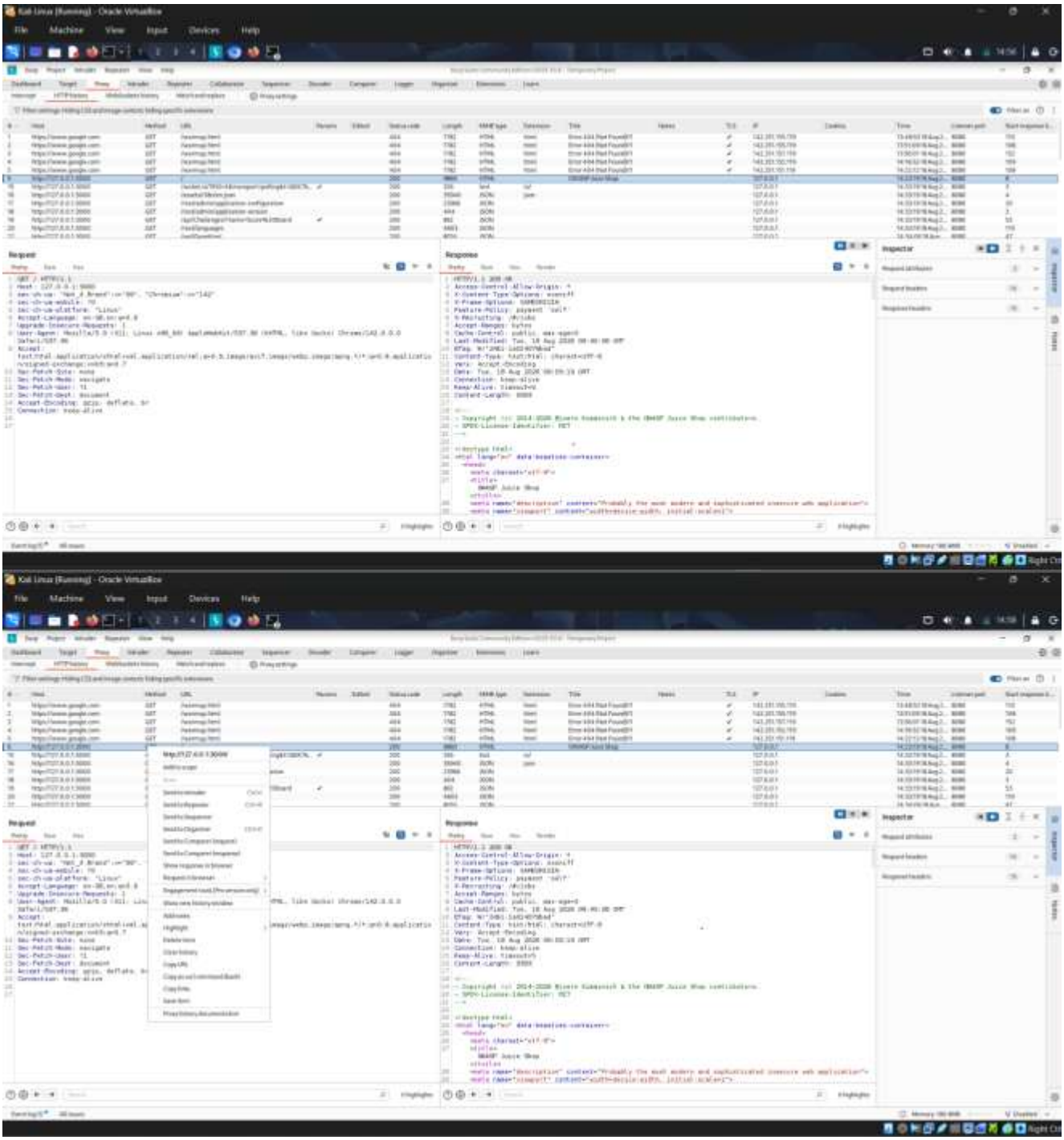
Important information includes:

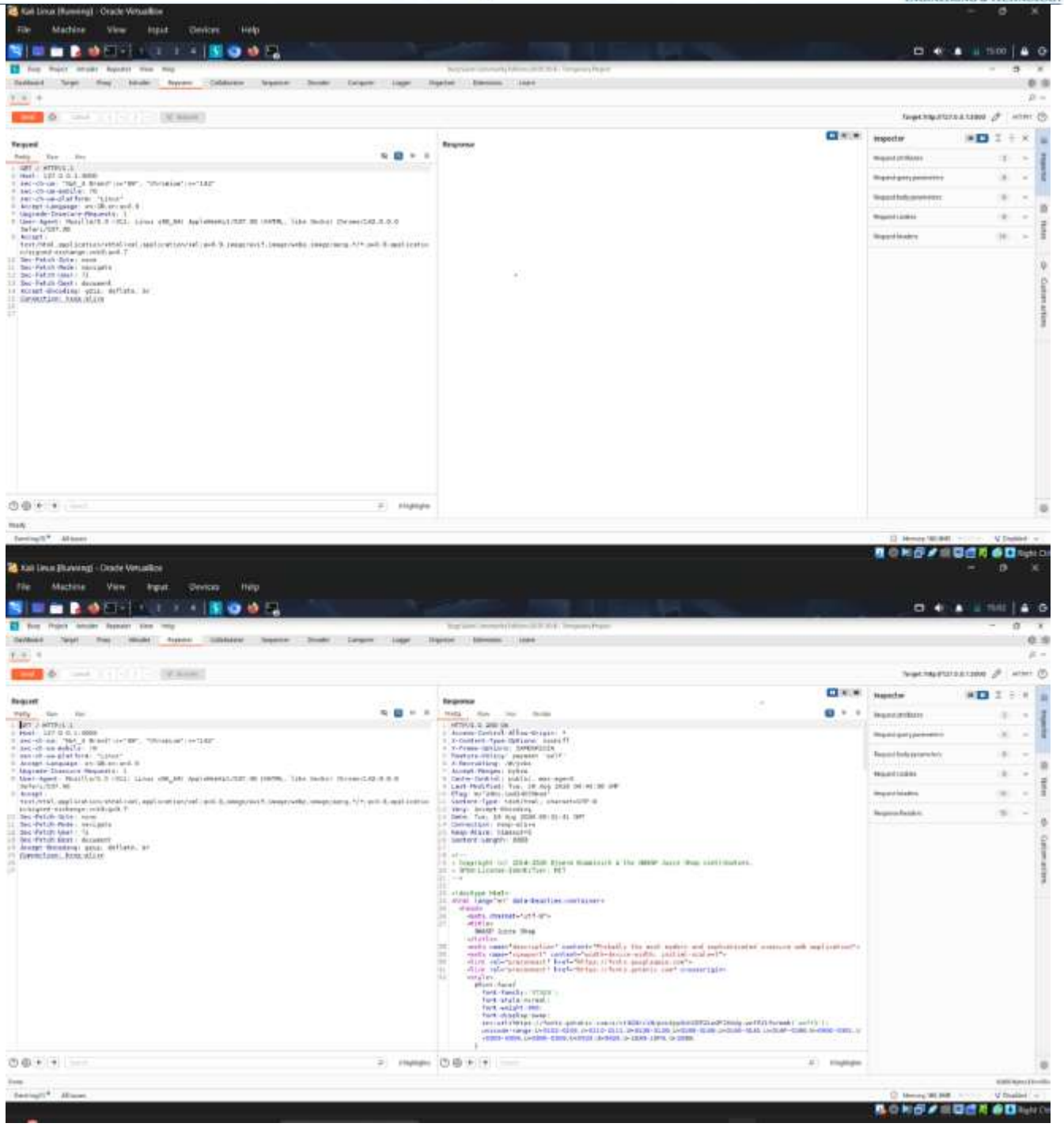
- HTTP status code
- Response length
- Response content
- Response time
- Differences between responses

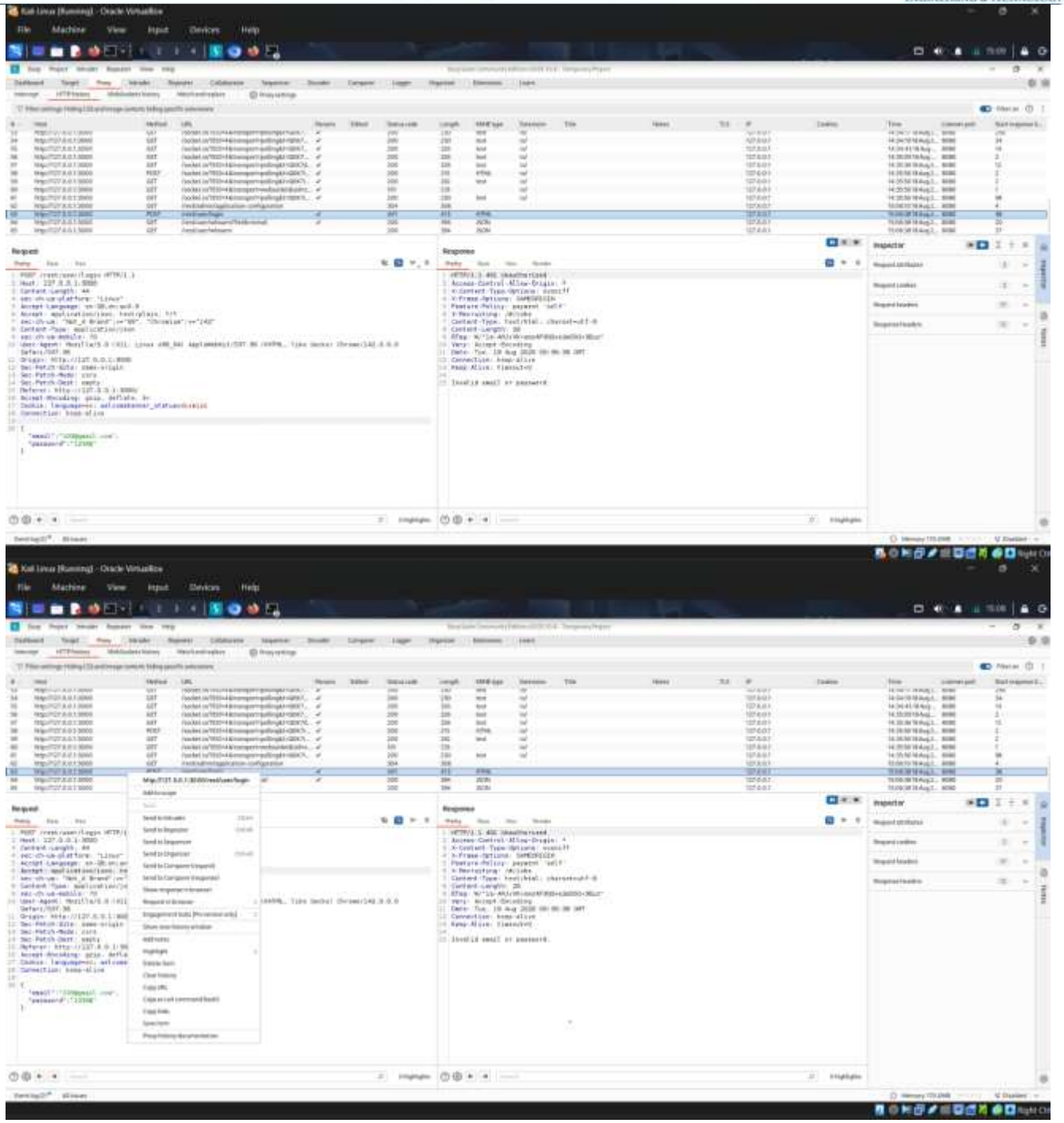
Expected Output:

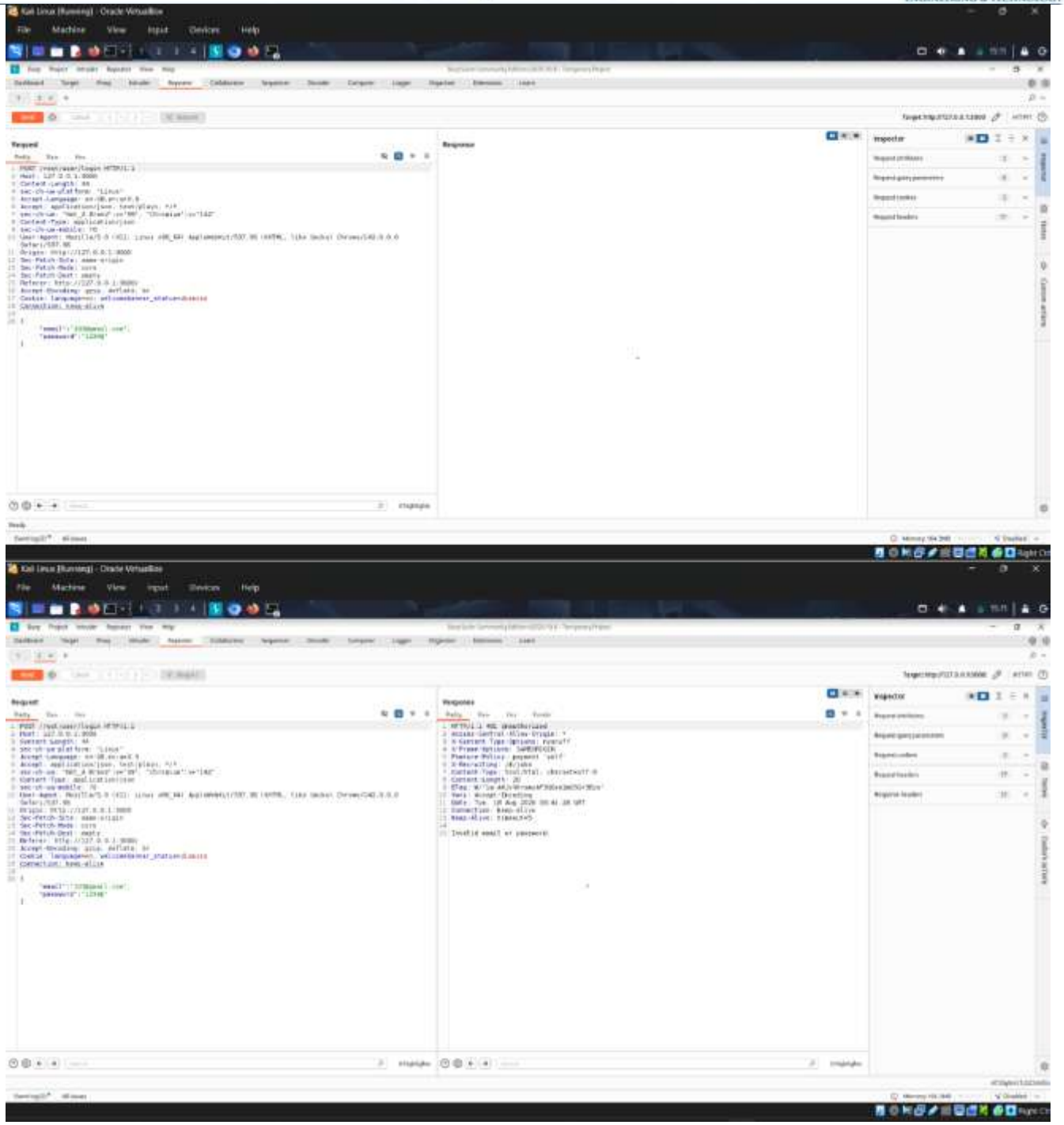
Different responses can be compared to identify unusual application behavior.

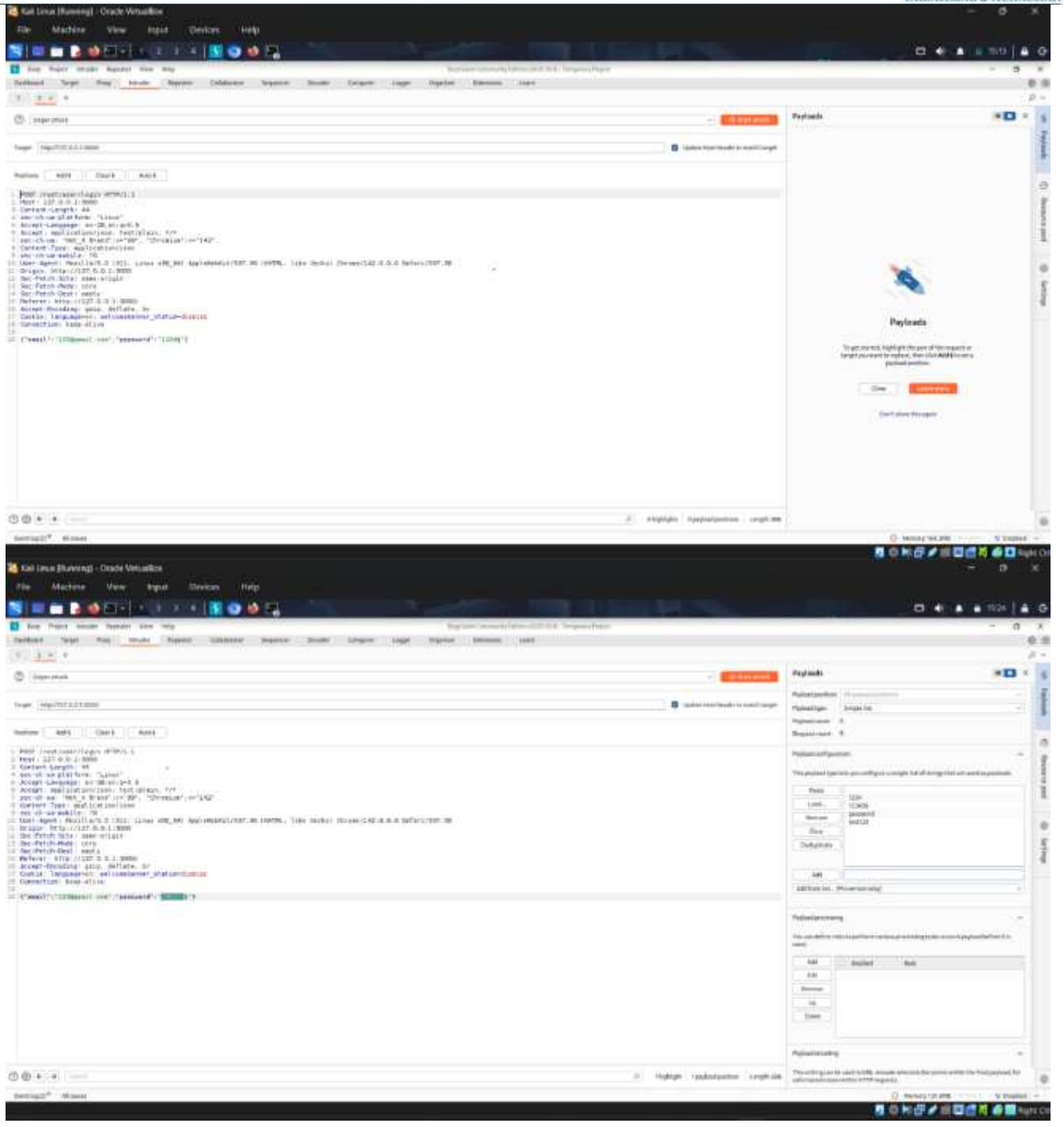
Outputs:

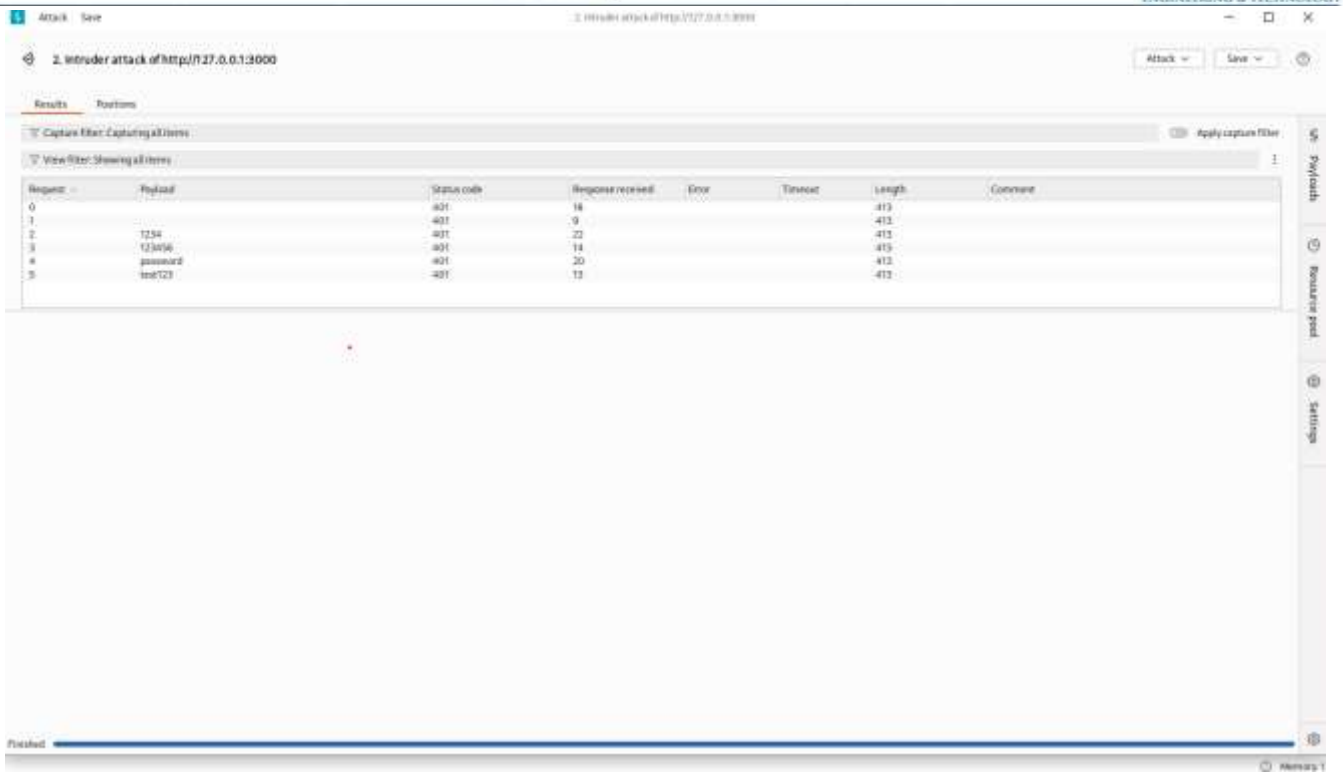





Kali Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Request
Response
Inspector


Kali Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Request
Response
Inspector


Kali Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Payloads



Observation:

The following observations are made:

1. HTTP requests can be captured using Burp Proxy.
2. Captured requests can be sent to Repeater.
3. Repeater allows individual requests to be modified and resent.
4. Server responses can be analyzed after each request.
5. Intruder can automate the testing of multiple input values.
6. Different responses can be compared based on status code, length, and content.
7. Repeater is suitable for manual testing, whereas Intruder is suitable for automated input testing.

Result and Discussion:

The experiment successfully demonstrated the use of **Burp Repeater and Burp Intruder**. Burp Repeater was used to manually modify and resend HTTP requests and analyze the resulting server responses. Burp Intruder was used to automate the testing of multiple authorized input values against a selected request parameter.

The experiment demonstrated how these tools can assist security testers in understanding application behavior and identifying potential weaknesses in input handling.

All testing should be performed only against systems and applications for which proper authorization has been obtained.

Conclusion:

This experiment demonstrated the practical use of Burp Repeater and Intruder for web application security testing. Repeater provides controlled manual testing of individual HTTP requests, while Intruder provides automated testing using multiple input values.

Both tools are useful for analyzing HTTP communication and testing the behavior of web applications in an authorized security testing environment.

Viva Questions:

1. What is Burp Repeater?
2. What is Burp Intruder?
3. What is the main difference between Repeater and Intruder?

For Faculty Use:

Correction Parameters	Formative Assessment [40%]	Timely completion of Practical [40%]	Attendance / Learning Attitude [20%]	
Marks Obtained				